MAT1375, Classwork16, Fall2026

Ch15. Exponential Equations and Applications

1. The Exponential and Logarithmic functions and one-to-one property:

For $b > 0, b \neq 1$, the exponential and logarithmic functions are one-to-one:

$b^x = b^y \Leftrightarrow$ $x = y$ (same output)

$\log_b(x) = \log_b(y) \Leftrightarrow$ $x = y$ (same input.)

2. Solve for x:

(a) $2^{x+7} = 32$.

$2^{x+7} = 2^5$

one-to-one property: $x+7 = 5$

$x = -2$

(b) $10^{2x-8} = 0.01$.

$0.01 = \frac{1}{100} = \frac{1}{10^2} = 10^{-2}$

$10^{2x-8} = 10^{-2}$

1-1 $\Rightarrow 2x - 8 = -2$

$x = 3$

(c) $27^{x+3} = 9^{x-1}$.

$27 = 3^3$; $9 = 3^2$

$(3^3)^{(x+3)} = (3^2)^{(x-1)}$

$3^{3(x+3)} = 3^{2(x-1)}$

1-1 $3(x+3) = 2(x-1)$

$3x + 9 = 2x - 2$

$\Rightarrow x = -11$

(d) $8^{x+2} = 4^{x-3}$.

$8 = 2^3$; $4 = 2^2$

$2^{3(x+2)} = 2^{2(x-3)}$

1-1 $3(x+2) = 2(x-3)$

$3x + 6 = 2x - 6$

$x = -12$.

3. How about the equations **without** the same base on both sides? (For example, $4^x = 15$.)

$4^x = 15 \Rightarrow x = \log_4(15)$

4. Using **Logarithms** to Solve **Exponential Equations**:

Step1: _____ the exponential expression.

Step2: Take _____ on both sides of the equation for common base _____ or natural base ____.

Step3: Simplify using one of these properties:

$\ln(b^x) =$_____if b is not 10 *or* $\ln(e^x) =$ _____ *or* $\log(10^x) =$ _____.

Step4: Solve for x.

5. Solve for x:

$$3^{x+5} = 8.$$

6. Solve for x.

(a) $13^{2x-4} = 6$. (b) $5.1^{x} = 2.7^{2x+6}$. (c) $7e^{2x} - 5 = 58$. (d) $10^{x} = 800$.